

Information and Passwords stealer – Practical II

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CAP6137 – Malware Reverse Engineering

Executive Summary:

This is an information and password stealer. Sample2.exe itself is an installer that installs the main 2 malicious payloads: wotsuper.exe and wotsuper2.exe. Sample 2 is packed using Bobsoft Delphi packer. They access multiple files and registries, and communicate with their C2 server. They contact multiple URLs, including ip-api.com and manillamemories.com. It is creating and accessing multiple DLLs. It is running queries to get passwords and system information, storing them in a txt, transmitting them to its C2, and deleting the zip and DLLs that it creates once it is done.

Static Analysis:

sections	8
compiler-stamp	0x2A425E19 (Fri Jun 19 18:22:17 1992)

The malware was compiled on **June 19, 1992** according to the compiler-stamp.

WINSZVersionValue	0x00000000
subsystem	GUI

It is a GUI program.

first-bytes-text	M Z P@
file-size	892326 (bytes)
size-without-overlay	181760 (bytes)
entropy	7.876
imphash	9623D7EC9B5866D2710C111879DB45FE
signature	BobSoft Mini Delphi -> BoB / BobSoft
entry-point	55 8B EC 83 C4 F0 B8 88 53 42 00 E8 24 F2 FD FF B8 C8 54 42 00 E8 2A 1C FE FF 8B 15 40 88 42 00 89
file-version	2.1
description	wotsuper 2.1 Installation

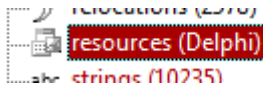
An overall entropy of 7.87 indicates that the program is likely packed. The file has a BobSoft signature, which is a common packing tool used to pack Delphi programs into an executable.

property	value
md5	86A248DC2BD0840445581D445640282F
sha1	54211CAFD97D19E282045D7CFFC58389FB24DEFB
sha256	0F4F4A5A03AD8144CB91257933AC8B246566CAF0C85664E7673E662F93FAB6...
entropy	7.999
file-offset	0x0002C600
size	710566 (bytes)
signature	Smart-installer
first-bytes-hex	53 6D 61 72 74 20 49 6E 73 74 61 6C 6C 20 4D 61 6B 65 72 20 76 2E 20 35 2E 30 ...
first-bytes-text	Smart Install Maker v. 5.04 ..0 ..0 ..
file-ratio	79.63 %

The 'Smart installer' overlay takes up 79.63% of the file, which is likely where the packed code is, because all other PE sections have <7 entropy and this section has an entropy of 7.999.

The sections are:

CODE: Stores the executable code. This is equivalent to .text sections, but is named as such here because of the use of Delphi, which names this section as 'CODE'.



DATA: Stores Global and Static variables. Equivalent to .data section, but same as CODE, is named as such because of the use of Delphi.

BSS: Used for allocating uninitialized Global and Static variables. This section only reserves memory for variables that will be initialized once the execution starts. It is sth to keep a close eye on when debugging.

.idata: Used for handling imported functions and libraries.

.tls: Used for defining thread local variables. The presence of this section is highly suspicious as it is commonly used for anti debugging techniques.

.rdata: Stores read only data.

.reloc: Contains base relocation instructions when the PE file does not get its preferred memory location.

.rsrc: Contains embedded resources, like icons, images, and sometimes, even malicious payload.

Suspicious imports:

FDIDestroy, FDICopy andFDICreate are used for handling cabinet archives. This suggests that the malware is an **installer**, and these functions could be used for installing the malicious payload.

The use of **GetKeyboardType** is very interesting.

```
uVar3 = 0;
iVar1 = GetKeyboardType(0);
if (iVar1 == 7) {
    uVar2 = GetKeyboardType(1);
    if (((uVar2 & 0xff00) == 0xd00) || ((uVar2 & 0xff00) == 0x400)) {
        uVar3 = 1;
    }
}
return uVar3;
```

This function is called in a function which is renamed for evaluating as 'checkkeyboard'

A screenshot of a debugger window showing a function call. The code is: `uVar2 = checkkeyboard();` followed by an if statement: `if ((char)uVar2 != '\0') {` and then `FUN_00402c44();`. The `checkkeyboard()` call is highlighted with a yellow box.

It runs a certain function only when the keyboard is a certain type. The 'certain function' in question accesses registry keys related to Delphi.

```

12 local_c = (uint)DAT_00426014;
13 LVar1 = RegOpenKeyExA((HKEY)0x80000002,"SOFTWARE\\Borland\\Delphi\\RTL",0,1,&local_8);
14 if (LVar1 == 0) {
15     uVar2 = *unaff_FS_OFFSET;
16     *unaff_FS_OFFSET = &stack0xffffffff4;
17     local_10 = 4;
18     RegQueryValueExA(local_8,"FPUMaskValue", (LPDWORD)0x0, (LPDWORD)0x0, (LPBYTE)&local_c,&local_10);
19     *unaff_FS_OFFSET = uVar2;
20     RegCloseKey(local_8);

```

Strings:

ascii	24	0x0002C838	-	url-pattern	-	http://www.wotsuper.com/
ascii	32	0x0002C920	-	url-pattern	-	https://iplogger.org/1Lda7.html
ascii	32	0x0002C9B0	-	url-pattern	-	https://iplogger.org/1smEq7.html

These are possibly URLs it tries to access.

ascii	52	0x00024EB8	-	utility	-	deflate 1.1.4 Copyright 1995-2002 Jean-loup Gailly
ascii	46	0x00025A58	-	utility	-	inflate 1.1.4 Copyright 1995-2002 Mark Adler

Mention of Adler and Gailly suggest the use of zlib and gzip.

ascii	24	0x0002C735	-	rtti	-	@\$%01\wotsuper\wotsuper
ascii	20	0x0002C80C	-	rtti	-	@\$%04\Uninstall.exe
ascii	20	0x0002C905	-	rtti	-	@\$%04\Uninstall.exe
ascii	20	0x0002C821	-	rtti	-	@\$%04\Uninstall.ini
ascii	20	0x0002C8D4	-	rtti	-	@\$%04\wotsuper1.exe
ascii	20	0x0002C967	-	rtti	-	@\$%04\wotsuper1.exe
ascii	19	0x0002C8ED	-	rtti	-	@\$%02\wotsuper.reg
ascii	19	0x0002C8BC	-	rtti	-	@\$%04\wotsuper.exe
ascii	19	0x0002C94A	-	rtti	-	@\$%04\wotsuper.exe

Hints at the creation or use of files called wotsuper.exe, wotsuper1.exe, uninstall.exe, uninstall.ini.

Dynamic Analysis:

Running sample2.exe immediately opened 2 internet explorer tabs, both with an error. This is because the malware is isolated from the real internet.

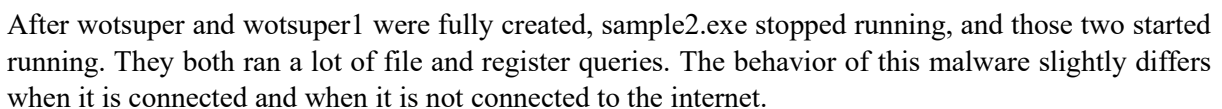
```

HKLM\SOFTWARE\Microsoft\Windows\Windows Error Reporting\TermReason\7316\Terminator: "HAM"
HKLM\SOFTWARE\Microsoft\Windows\Windows Error Reporting\TermReason\7316\Reason: 0x00000004
HKLM\SOFTWARE\Microsoft\Windows\Windows Error Reporting\TermReason\7316\CreationTime: 4F 2F 19 A5 C1 98 DB 01
HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\DisplayName: "wotsuper 2.1"
HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\DisplayVersion: "2.1"
HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\VersionMajor: 0x00000002
HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\VersionMinor: 0x00000001
HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\Publisher: "wotsuper"
HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\DisplayIcon: "C:\Program Files (x86)\wotsuper\wotsuper\Uninstall.exe"
HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\UninstallString: "C:\Program Files (x86)\wotsuper\wotsuper\Uninstall.exe"
HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\URLInfoAbout: "http://www.wotsuper.com/"
HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\HelpLink: "wotsuper@company.com"
HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\InstallLocation: "C:\Program Files (x86)\wotsuper\wotsuper\"
HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\InstallSource: "C:\Users\Malware\Desktop\"
HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\InstallDate: "20250319"
HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\Language: 0x00000419
HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\EstimatedSize: 0x00000444
HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\NoModify: 0x00000001
HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Uninstall\wotsuper 2.1\NoRepair: 0x00000001

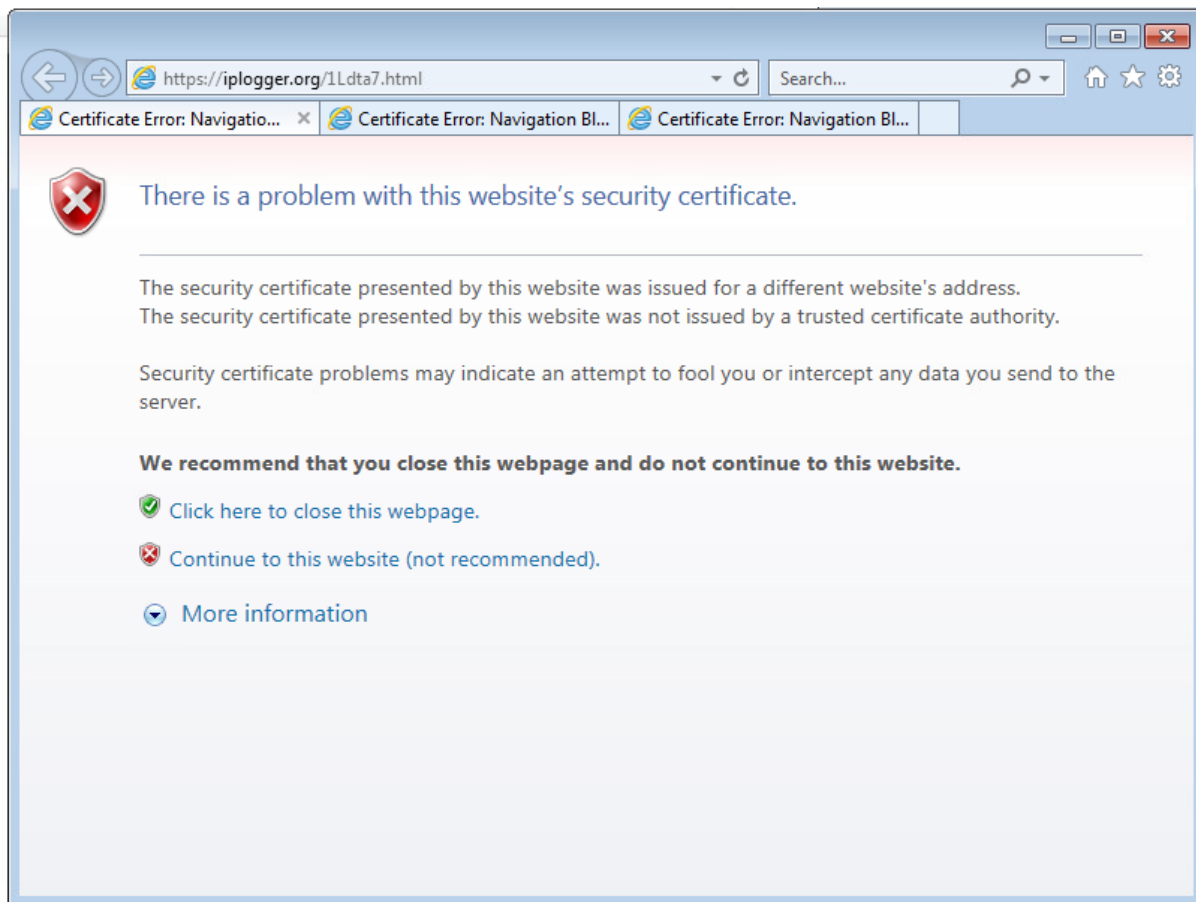
```

Registry changes related to newly created wotsuper were observed. Other registry changes were mainly related to internet explorer.

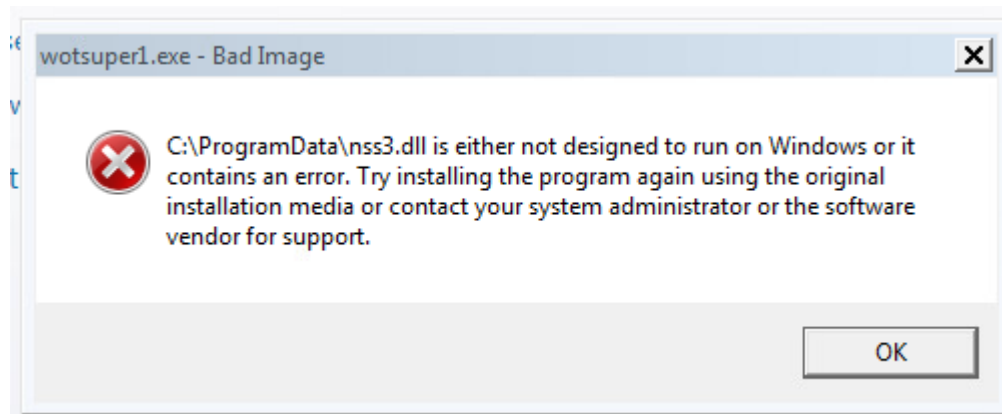
sample2.exe did a LOT of ReadFile operations on itself, and eventually created and wrote to C:\Program Files\wotsuper\ (Uninstall.exe/Uninstall.ini/wotsuper.exe/wotsuper1.exe).



ONLINE (Connected to the internet) EXECUTION:



An internet connection was simulated by running fakedns, inetsim, and accept-all-ips. The error that the website displays is a certificate error, which is expected with inetsim. Wotsuper1 threw the following error.



Note that nss3.dll provides multiple encryption-decryption functionality and is developed by Mozilla.

Pressing 'OK' multiple times would eventually stop the execution of wotsuper1. Before pressing ok for the final time, these dll files can be observed in the ProgramData folder:

Name	Date modified	Type	Size
freebl3.dll	3/28/2025 8:36 PM	Application extens...	1 KB
mozglue.dll	3/28/2025 8:36 PM	Application extens...	1 KB
msvcpl140.dll	3/28/2025 8:36 PM	Application extens...	1 KB
nss3.dll	3/28/2025 8:36 PM	Application extens...	1 KB
softoken3.dll	3/28/2025 8:36 PM	Application extens...	1 KB
vcruntime140.dll	3/28/2025 8:36 PM	Application extens...	1 KB
13E4TW9JH8592SFND9WS0LU24	3/28/2025 8:36 PM	File folder	
G9IAC7CRBDKPZOEQ1XLAWPAQ4	3/28/2025 8:30 PM	File folder	
Package Cache	1/25/2024 1:30 PM	File folder	
Microsoft	3/3/2022 10:12 AM	File folder	
VMware	3/2/2022 6:50 PM	File folder	

The folders with seemingly gibberish names are also created by Wotsuper1. This folder's name is different every time.

11:14:...		wotsuper1.exe	2836		ReadFile	C:\\$Directory
11:14:...		wotsuper1.exe	2836		WriteFile	C:\ProgramData\WDGRKFXG83YQ0POCSL6S7F7K2\files\information.txt
11:14:...		wotsuper1.exe	2836		ReadFile	C:\ProgramData\WDGRKFXG83YQ0POCSL6S7F7K2\files\information.txt
11:14:...		wotsuper1.exe	2836		ReadFile	C:\ProgramData\WDGRKFXG83YQ0POCSL6S7F7K2\files\information.txt
11:14:...		wotsuper1.exe	2836		WriteFile	C:\ProgramData\WDGRKFXG83YQ0POCSL6S7F7K2_63c09915-9fbb-48a0-b874f23fc25b54d58865967895.zip
11:14:...		wotsuper1.exe	2836		WriteFile	C:\ProgramData\WDGRKFXG83YQ0POCSL6S7F7K2_63c09915-9fbb-48a0-b874f23fc25b54d58865967895.zip
11:14:...		wotsuper1.exe	2836		WriteFile	C:\ProgramData\WDGRKFXG83YQ0POCSL6S7F7K2_63c09915-9fbb-48a0-b874f23fc25b54d58865967895.zip
11:14:...		wotsuper1.exe	2836		WriteFile	C:\ProgramData\WDGRKFXG83YQ0POCSL6S7F7K2_63c09915-9fbb-48a0-b874f23fc25b54d58865967895.zip
11:14:...		wotsuper1.exe	2836		WriteFile	C:\ProgramData\WDGRKFXG83YQ0POCSL6S7F7K2_63c09915-9fbb-48a0-b874f23fc25b54d58865967895.zip
11:14:...		wotsuper1.exe	2836		WriteFile	C:\ProgramData\WDGRKFXG83YQ0POCSL6S7F7K2_63c09915-9fbb-48a0-b874f23fc25b54d58865967895.zip
11:14:...		wotsuper1.exe	2836		WriteFile	C:\ProgramData\WDGRKFXG83YQ0POCSL6S7F7K2_63c09915-9fbb-48a0-b874f23fc25b54d58865967895.zip
11:14:...		wotsuper1.exe	2836		CreateFile	C:\ProgramData\WDGRKFXG83YQ0POCSL6S7F7K2\files\passwords.txt
11:14:...		wotsuper1.exe	2836		CloseFile	C:\ProgramData\WDGRKFXG83YQ0POCSL6S7F7K2\files\passwords.txt
11:14:...		wotsuper1.exe	2836		CreateFile	C:\ProgramData\WDGRKFXG83YQ0POCSL6S7F7K2\files\passwords.txt
11:14:...		wotsuper1.exe	2836		QueryStandardIcon	C:\ProgramData\WDGRKFXG83YQ0POCSL6S7F7K2\files\passwords.txt
11:14:...		wotsuper1.exe	2836		CloseFile	C:\ProgramData\WDGRKFXG83YQ0POCSL6S7F7K2\files\passwords.txt
11:14:...		wotsuper1.exe	2836		CreateFile	C:\ProgramData\WDGRKFXG83YQ0POCSL6S7F7K2\files\passwords.txt
11:14:...		wotsuper1.exe	2836		QueryStandardIcon	C:\ProgramData\WDGRKFXG83YQ0POCSL6S7F7K2\files\passwords.txt
11:14:...		wotsuper1.exe	2836		CloseFile	C:\ProgramData\WDGRKFXG83YQ0POCSL6S7F7K2\files\passwords.txt
11:14:...		wotsuper1.exe	2836		CreateFile	C:\ProgramData\WDGRKFXG83YQ0POCSL6S7F7K2\files\passwords.txt
11:14:...		wotsuper1.exe	2836		QueryStandardIcon	C:\ProgramData\WDGRKFXG83YQ0POCSL6S7F7K2\files\passwords.txt
11:14:...		wotsuper1.exe	2836		CloseFile	C:\ProgramData\WDGRKFXG83YQ0POCSL6S7F7K2\files\passwords.txt
11:14:...		wotsuper1.exe	2836		CreateFile	C:\ProgramData\WDGRKFXG83YQ0POCSL6S7F7K2\files\passwords.txt

The above screenshot shows the creation of this folder, and also some 'information.txt' and 'passwords.txt' in this folder. (Note: This is from a different run of the malware, so the folder name differs from the one in file explorer). These are some of the register and folder queries wotsuper and wotsuper1 run:

10:47:...	wotsuper1.exe	3544	RegOpenKey	HKLM\System\CurrentControlSet\Control\Session Manager
10:47:...	wotsuper1.exe	3544	RegOpenKey	HKLM\System\CurrentControlSet\Control\Session Manager
10:47:...	wotsuper1.exe	3544	RegQueryValue	HKLM\System\CurrentControlSet\Control\Session Manager\CWDIllegalInDLLSearch
10:47:...	wotsuper1.exe	3544	RegCloseKey	HKLM\System\CurrentControlSet\Control\Session Manager
10:47:...	wotsuper1.exe	3544	RegOpenKey	HKLM\System\CurrentControlSet\Control\SafeBoot\Option
10:47:...	wotsuper1.exe	3544	RegOpenKey	HKLM\System\CurrentControlSet\Control\SafeBoot\Option
10:47:...	wotsuper1.exe	3544	RegOpenKey	HKLM\System\CurrentControlSet\Control\Srp\GP\DLL
10:47:...	wotsuper1.exe	3544	RegOpenKey	HKLM\System\CurrentControlSet\Control\Srp\GP\DLL
10:47:...	wotsuper1.exe	3544	RegOpenKey	HKLM\Software\Policies\Microsoft\Windows\Safer\CodeIdentifiers
10:47:...	wotsuper1.exe	3544	RegQueryValue	HKLM\SOFTWARE\Policies\Microsoft\Windows\safer\codeidentifiers\TransparentEnabled
10:47:...	wotsuper1.exe	3544	RegCloseKey	HKLM\SOFTWARE\Policies\Microsoft\Windows\safer\codeidentifiers
10:47:...	wotsuper1.exe	3544	RegOpenKey	HKCU\Software\Policies\Microsoft\Windows\Safer\CodeIdentifiers
10:47:...	wotsuper1.exe	3544	RegOpenKey	HKLM\Software\Microsoft\Windows\CurrentVersion\SideBySide\AssemblyStorageRoots
10:47:...	wotsuper1.exe	3544	RegOpenKey	HKLM\System\CurrentControlSet\Control\Nls\Sorting\Versions
10:47:...	wotsuper1.exe	3544	RegOpenKey	HKLM\System\CurrentControlSet\Control\Nls\Sorting\Versions
10:47:...	wotsuper1.exe	3544	RegQueryValue	HKLM\System\CurrentControlSet\Control\Nls\Sorting\Versions\CompatDll
10:47:...	wotsuper1.exe	3544	RegQueryValue	HKLM\System\CurrentControlSet\Control\Nls\Sorting\Versions\00060101.00060101
10:47:...	wotsuper1.exe	3544	RegQueryValue	HKLM\System\CurrentControlSet\Control\Nls\Sorting\Versions(Default)
10:47:...	wotsuper1.exe	3544	RegOpenKey	HKLM\System\CurrentControlSet\Control\Session Manager
10:47:...	wotsuper1.exe	3544	RegOpenKey	HKLM\System\CurrentControlSet\Control\Session Manager
10:47:...	wotsuper1.exe	3544	RegQueryValue	HKLM\System\CurrentControlSet\Control\Session Manager\SafeDllSearchMode
10:47:...	wotsuper1.exe	3544	RegOpenKey	HKLM\System\CurrentControlSet\Control\Error Message Instrument\
10:47:...	wotsuper1.exe	3544	RegOpenKey	HKLM\System\CurrentControlSet\Control\Error Message Instrument
10:47:...	wotsuper1.exe	3544	RegOpenKey	HKLM\Software\Microsoft\Windows NT\CurrentVersion\GRE_Initialize
10:47:...	wotsuper1.exe	3544	RegQueryValue	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\GRE_Initialize\DisableMetaFiles
10:47:...	wotsuper1.exe	3544	RegCloseKey	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\GRE_Initialize
10:47:...	wotsuper1.exe	3544	RegOpenKey	HKLM\Software\Microsoft\Windows NT\CurrentVersion\Compatibility32
10:47:...	wotsuper1.exe	704	RegQueryValue	HKLM\System\CurrentControlSet\Services\Tcpip\Parameters\Winsock\UseDelayedAcceptance
10:47:...	wotsuper.exe	704	RegQueryValue	HKLM\System\CurrentControlSet\services\TCP\Parameters\Winsock\HelperDllName
10:47:...	wotsuper.exe	704	RegCloseKey	HKLM\System\CurrentControlSet\services\TCP\Parameters\Winsock
10:47:...	wotsuper.exe	704	RegCreateKey	HKLM\System\CurrentControlSet\Services\Tcpip\Parameters
10:47:...	wotsuper.exe	704	RegCreateKey	HKLM\System\CurrentControlSet\Services\Tcpip\Parameters
10:47:...	wotsuper.exe	704	RegOpenKey	HKLM\System\CurrentControlSet\Services\DnsCache\Parameters
10:47:...	wotsuper.exe	704	RegOpenKey	HKLM\System\CurrentControlSet\Services\DnsCache\Parameters
10:47:...	wotsuper.exe	704	RegOpenKey	HKLM\Software\Policies\Microsoft\Windows NT\DnsClient
10:47:...	wotsuper.exe	704	RegQueryValue	HKLM\System\CurrentControlSet\services\Tcpip\Parameters\Hostname
10:47:...	wotsuper.exe	704	RegQueryValue	HKLM\System\CurrentControlSet\services\Tcpip\Parameters\Hostname
10:47:...	wotsuper.exe	704	RegCloseKey	HKLM\System\CurrentControlSet\services\Tcpip\Parameters
10:47:...	wotsuper.exe	704	RegCloseKey	HKLM\System\CurrentControlSet\services\DnsCache\Parameters
10:47:...	wotsuper.exe	704	RegCreateKey	HKLM\System\CurrentControlSet\Services\Tcpip\Parameters
10:47:...	wotsuper.exe	704	RegCreateKey	HKLM\System\CurrentControlSet\Services\Tcpip\Parameters
10:47:...	wotsuper.exe	704	RegOpenKey	HKLM\System\CurrentControlSet\Services\DnsCache\Parameters
10:47:...	wotsuper.exe	704	RegOpenKey	HKLM\System\CurrentControlSet\Services\DnsCache\Parameters
10:47:...	wotsuper.exe	704	RegOpenKey	HKLM\Software\Policies\Microsoft\Windows NT\DnsClient
10:47:...	wotsuper.exe	704	RegQueryValue	HKLM\System\CurrentControlSet\services\Tcpip\Parameters\Hostname
10:47:...	wotsuper.exe	704	RegQueryValue	HKLM\System\CurrentControlSet\services\Tcpip\Parameters\Hostname
10:47:...	wotsuper.exe	704	RegCloseKey	HKLM\System\CurrentControlSet\services\Tcpip\Parameters
10:47:...	wotsuper.exe	704	RegCloseKey	HKLM\System\CurrentControlSet\services\DnsCache\Parameters
10:47:...	wotsuper.exe	704	RegCreateKey	HKLM\System\CurrentControlSet\Services\Tcpip\Parameters
10:47:...	wotsuper1.exe	3544	CloseFile	C:\Users\malware\AppData\Roaming
10:47:...	wotsuper1.exe	3544	CreateFile	C:\Users\malware\AppData\Roaming\Microsoft\Windows\Cookies
10:47:...	wotsuper1.exe	3544	CreateFile	C:\Users\malware\AppData\Roaming\Microsoft\Windows\Cookies
10:47:...	wotsuper1.exe	3544	QueryBasicInfor...	C:\Users\malware\AppData\Roaming\Microsoft\Windows\Cookies
10:47:...	wotsuper1.exe	3544	CloseFile	C:\Users\malware\AppData\Roaming\Microsoft\Windows\Cookies
10:47:...	wotsuper1.exe	3544	CreateFile	C:\Users\malware\AppData\Roaming\Microsoft\Windows\Cookies
10:47:...	wotsuper1.exe	3544	QueryBasicInfor...	C:\Users\malware\AppData\Roaming\Microsoft\Windows\Cookies
10:47:...	wotsuper1.exe	3544	CloseFile	C:\Users\malware\AppData\Roaming\Microsoft\Windows\Cookies
10:47:...	wotsuper1.exe	3544	CreateFile	C:\Users\malware\AppData\Roaming\Microsoft\Windows\Cookies
10:47:...	wotsuper1.exe	3544	CreateFile	C:\Users\malware
10:47:...	wotsuper1.exe	3544	QueryBasicInfor...	C:\Users\malware
10:47:...	wotsuper1.exe	3544	CloseFile	C:\Users\malware
10:47:...	wotsuper1.exe	3544	CreateFile	C:\Users\malware\AppData\Local
10:47:...	wotsuper1.exe	3544	CreateFile	C:\Users\malware\AppData\Local
10:47:...	wotsuper1.exe	3544	QueryBasicInfor...	C:\Users\malware\AppData\Local
10:47:...	wotsuper1.exe	3544	CloseFile	C:\Users\malware\AppData\Local
10:47:...	wotsuper1.exe	3544	CreateFile	C:\Users\malware\AppData\Local\Microsoft\Windows\History
10:47:...	wotsuper1.exe	3544	CreateFile	C:\Users\malware\AppData\Local\Microsoft\Windows\History
10:47:...	wotsuper1.exe	3544	QueryBasicInfor...	C:\Users\malware\AppData\Local\Microsoft\Windows\History
10:47:...	wotsuper1.exe	3544	CloseFile	C:\Users\malware\AppData\Local\Microsoft\Windows\History
10:47:...	wotsuper1.exe	3544	CreateFile	C:\Users\malware\AppData\Local\Microsoft\Windows\History\History.IE5
10:47:...	wotsuper1.exe	3544	QueryBasicInfor...	C:\Users\malware\AppData\Local\Microsoft\Windows\History\History.IE5
10:47:...	wotsuper1.exe	3544	CloseFile	C:\Users\malware\AppData\Local\Microsoft\Windows\History\History.IE5
10:47:...	wotsuper1.exe	3544	CreateFile	C:\Windows\Registration\R000000000001e.clb
10:47:...	wotsuper1.exe	3544	QueryStandard...	C:\Windows\Registration\R000000000001e.clb
10:47:...	wotsuper1.exe	3544	CreateFileMapp...	C:\Windows\Registration\R000000000001e.clb
10:47:...	wotsuper1.exe	3544	QueryStandard...	C:\Windows\Registration\R000000000001e.clb
10:47:...	wotsuper1.exe	3544	CreateFileMapp...	C:\Windows\Registration\R000000000001e.clb
10:47:...	wotsuper1.exe	3544	CreateFile	C:\Program Files\wotsuper\wotsuper\wotsuper1.exe.Local
10:47:...	wotsuper1.exe	3544	CreateFile	C:\Windows\winsxs\x86_microsoft.windows.common-controls_6595b64144ccf1df_6.0.7601.18837_none_41e855142bd5705d
10:47:...	wotsuper1.exe	3544	QueryBasicInfor...	C:\Windows\winsxs\x86_microsoft.windows.common-controls_6595b64144ccf1df_6.0.7601.18837_none_41e855142bd5705d
10:47:...	wotsuper1.exe	3544	CloseFile	C:\Windows\winsxs\x86_microsoft.windows.common-controls_6595b64144ccf1df_6.0.7601.18837_none_41e855142bd5705d
10:47:...	wotsuper1.exe	3544	CreateFile	C:\Windows\winsxs\x86_microsoft.windows.common-controls_6595b64144ccf1df_6.0.7601.18837_none_41e855142bd5705d
10:47:...	wotsuper1.exe	3544	CreateFile	C:\Windows\WindowsShell.Manifest

10:47....	wotsuper.exe	704	CloseFile	C:\Windows\System32\bcrypt.dll
10:47....	wotsuper.exe	704	CreateFile	C:\Program Files\wotsuper\wotsuper\WINHTTP.dll
10:47....	wotsuper.exe	704	CreateFile	C:\Windows\System32\winhttp.dll
10:47....	wotsuper.exe	704	QueryBasicInfor...	C:\Windows\System32\winhttp.dll
10:47....	wotsuper.exe	704	CloseFile	C:\Windows\System32\winhttp.dll
10:47....	wotsuper.exe	704	CreateFile	C:\Windows\System32\winhttp.dll
10:47....	wotsuper.exe	704	CreateFileMapp...	C:\Windows\System32\winhttp.dll
10:47....	wotsuper.exe	704	CreateFileMapp...	C:\Windows\System32\winhttp.dll
10:47....	wotsuper.exe	704	CloseFile	C:\Windows\System32\winhttp.dll
10:47....	wotsuper.exe	704	CreateFile	C:\Program Files\wotsuper\wotsuper\webio.dll
10:47....	wotsuper.exe	704	CreateFile	C:\Windows\System32\webio.dll
10:47....	wotsuper.exe	704	QueryBasicInfor...	C:\Windows\System32\webio.dll
10:47....	wotsuper.exe	704	CloseFile	C:\Windows\System32\webio.dll
10:47....	wotsuper.exe	704	CreateFile	C:\Windows\System32\webio.dll
10:47....	wotsuper.exe	704	CreateFileMapp...	C:\Windows\System32\webio.dll
10:47....	wotsuper.exe	704	CreateFileMapp...	C:\Windows\System32\webio.dll
10:47....	wotsuper.exe	704	CloseFile	C:\Windows\System32\webio.dll
10:47....	wotsuper.exe	704	CreateFile	C:\Program Files\wotsuper\wotsuper\wotsuper.exe.Local
10:47....	wotsuper.exe	704	CreateFile	C:\Windows\winsxs\x86_microsoft.windows.gdiplus_6595b64144ccf1df_1.1.7601.24203_none_5c030043a0118bf
10:47....	wotsuper.exe	704	QueryBasicInfor...	C:\Windows\winsxs\x86_microsoft.windows.gdiplus_6595b64144ccf1df_1.1.7601.24203_none_5c030043a0118bf
10:47....	wotsuper.exe	704	CloseFile	C:\Windows\winsxs\x86_microsoft.windows.gdiplus_6595b64144ccf1df_1.1.7601.24203_none_5c030043a0118bf
10:47....	wotsuper.exe	704	CreateFile	C:\Windows\winsxs\x86_microsoft.windows.gdiplus_6595b64144ccf1df_1.1.7601.24203_none_5c030043a0118bf
10:47....	wotsuper.exe	704	CreateFile	C:\Windows\winsxs\x86_microsoft.windows.gdiplus_6595b64144ccf1df_1.1.7601.24203_none_5c030043a0118bf\GdiPlus.dll
10:47....	wotsuper.exe	704	QueryBasicInfor...	C:\Windows\winsxs\x86_microsoft.windows.gdiplus_6595b64144ccf1df_1.1.7601.24203_none_5c030043a0118bf\GdiPlus.dll
10:47....	wotsuper.exe	704	CloseFile	C:\Windows\winsxs\x86_microsoft.windows.gdiplus_6595b64144ccf1df_1.1.7601.24203_none_5c030043a0118bf\GdiPlus.dll
10:47....	wotsuper.exe	704	CreateFile	C:\Windows\winsxs\x86_microsoft.windows.gdiplus_6595b64144ccf1df_1.1.7601.24203_none_5c030043a0118bf\GdiPlus.dll
10:47....	wotsuper.exe	704	CreateFileMapp...	C:\Windows\winsxs\x86_microsoft.windows.gdiplus_6595b64144ccf1df_1.1.7601.24203_none_5c030043a0118bf\GdiPlus.dll
10:47....	wotsuper.exe	704	CreateFileMapp...	C:\Windows\winsxs\x86_microsoft.windows.gdiplus_6595b64144ccf1df_1.1.7601.24203_none_5c030043a0118bf\GdiPlus.dll
10:47....	wotsuper.exe	704	CloseFile	C:\Windows\winsxs\x86_microsoft.windows.gdiplus_6595b64144ccf1df_1.1.7601.24203_none_5c030043a0118bf\GdiPlus.dll
10:47....	wotsuper.exe	704	CreateFile	C:\Program Files\wotsuper\wotsuper\msvcr100.dll
10:47....	wotsuper.exe	704	CreateFile	C:\Windows\System32\msvcr100.dll
10:47....	wotsuper.exe	704	CreateFile	C:\Windows\System32\msvcr100.dll
10:47....	wotsuper.exe	704	CreateFile	C:\Windows\System32\msvcr100.dll
10:47....	wotsuper.exe	704	CreateFile	C:\Program Files\wotsuper\wotsuper\msvcr100.dll
10:47....	wotsuper.exe	704	CreateFile	C:\Windows\System32\msvcr100.dll

A lot of system info registries and folders can be seen getting accessed.

Wotsuper1 created zip files with seemingly gibberish names. Upon unzipping one of them, an “information.txt” and “passwords.txt” were found.

10:47....	wotsuper1.exe	3544	RegOpenKey	HKLM\SOFTWARE\Microsoft\Internet Explorer\MAIN\FeatureControl\FEATURE_HTTP_USERNAME_PASSWORD_DISABLE
10:47....	wotsuper1.exe	3544	RegQueryValue	HKLM\SOFTWARE\Microsoft\Internet Explorer\MAIN\FeatureControl\FEATURE_HTTP_USERNAME_PASSWORD_DISABLE\wotsuper1.exe
10:47....	wotsuper1.exe	3544	RegQueryValue	HKLM\SOFTWARE\Microsoft\Internet Explorer\MAIN\FeatureControl\FEATURE_HTTP_USERNAME_PASSWORD_DISABLE*
10:47....	wotsuper1.exe	3544	RegCloseKey	HKLM\SOFTWARE\Microsoft\Internet Explorer\MAIN\FeatureControl\FEATURE_HTTP_USERNAME_PASSWORD_DISABLE
10:47....	wotsuper1.exe	3544	CreateFile	C:\Program Data\GDI\HIL\PIPI_71XU117C11F91Y0f0\file\wotsuper1.txt

It can also be seen accessing Internet Explorer passwords through registries.

10:47....	wotsuper1.exe	3544	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Services\crypt32
10:47....	wotsuper1.exe	3544	RegOpenKey	HKLM\System\CurrentControlSet\Services\crypt32
10:47....	wotsuper1.exe	3544	RegQueryValue	HKLM\System\CurrentControlSet\Services\crypt32\DebugHeapFlags
10:47....	wotsuper1.exe	3544	RegCloseKey	HKLM\System\CurrentControlSet\Services\crypt32
10:47....	wotsuper1.exe	3544	RegOpenKey	HKLM\SOFTWARE\Microsoft\Cryptography
10:47....	wotsuper1.exe	3544	RegQueryValue	HKLM\SOFTWARE\Microsoft\Cryptography\MachineGuid
10:47....	wotsuper1.exe	3544	RegQueryValue	HKLM\SOFTWARE\Microsoft\Cryptography\MachineGuid
10:47....	wotsuper1.exe	3544	RegCloseKey	HKLM\SOFTWARE\Microsoft\Cryptography
10:47....	wotsuper1.exe	3544	RegOpenKey	HKLM\SOFTWARE\Microsoft\Internet Explorer\MAIN\FeatureControl\FEATURE_HTTP_USERNAME_PASSWORD_DISABLE
10:47....	wotsuper1.exe	3544	RegQueryValue	HKLM\SOFTWARE\Microsoft\Internet Explorer\MAIN\FeatureControl\FEATURE_HTTP_USERNAME_PASSWORD_DISABLE\wotsuper1.exe
10:47....	wotsuper1.exe	3544	RegQueryValue	HKLM\SOFTWARE\Microsoft\Internet Explorer\MAIN\FeatureControl\FEATURE_HTTP_USERNAME_PASSWORD_DISABLE*
10:47....	wotsuper1.exe	3544	RegCloseKey	HKLM\SOFTWARE\Microsoft\Internet Explorer\MAIN\FeatureControl\FEATURE_HTTP_USERNAME_PASSWORD_DISABLE
10:47....	wotsuper1.exe	3544	RegOpenKey	HKLM\SOFTWARE\Microsoft\Cryptography\Defaults\Provider\Microsoft Strong Cryptographic Provider
10:47....	wotsuper1.exe	3544	RegQueryValue	HKLM\SOFTWARE\Microsoft\Cryptography\Defaults\Provider\Microsoft Strong Cryptographic Provider\Type
10:47....	wotsuper1.exe	3544	RegQueryValue	HKLM\SOFTWARE\Microsoft\Cryptography\Defaults\Provider\Microsoft Strong Cryptographic Provider\Image Path
10:47....	wotsuper1.exe	3544	RegQueryValue	HKLM\SOFTWARE\Microsoft\Cryptography\Defaults\Provider\Microsoft Strong Cryptographic Provider\Image Path
10:47....	wotsuper1.exe	3544	RegQueryValue	HKLM\SOFTWARE\Microsoft\Cryptography\Defaults\Provider\Microsoft Strong Cryptographic Provider\Image Path
10:47....	wotsuper1.exe	3544	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Policies\Microsoft\Cryptography\Configuration
10:47....	wotsuper1.exe	3544	RegOpenKey	HKLM\System\CurrentControlSet\Policies\Microsoft\Cryptography\Configuration
10:47....	wotsuper1.exe	3544	RegOpenKey	HKLM\Software\Policies\Microsoft\Cryptography
10:47....	wotsuper1.exe	3544	RegQueryValue	HKLM\SOFTWARE\Policies\Microsoft\Cryptography\PrivKeyCacheMaxItems
10:47....	wotsuper1.exe	3544	RegQueryValue	HKLM\SOFTWARE\Policies\Microsoft\Cryptography\PrivKeyCachePurgeIntervalSeconds
10:47....	wotsuper1.exe	3544	RegQueryValue	HKLM\SOFTWARE\Policies\Microsoft\Cryptography\PrivateKeyLifetimeSeconds
10:47....	wotsuper1.exe	3544	RegCloseKey	HKLM\SOFTWARE\Policies\Microsoft\Cryptography
10:47....	wotsuper1.exe	3544	RegOpenKey	HKLM\Software\Microsoft\Cryptography
10:47....	wotsuper1.exe	3544	RegQueryValue	HKLM\SOFTWARE\Microsoft\Cryptography\MachineGuid
10:47....	wotsuper1.exe	3544	RegQueryValue	HKLM\SOFTWARE\Microsoft\Cryptography\MachineGuid
10:47....	wotsuper1.exe	3544	RegQueryValue	HKLM\SOFTWARE\Microsoft\Cryptography\MachineGuid
10:47....	wotsuper1.exe	3544	RegQueryValue	HKLM\SOFTWARE\Microsoft\Cryptography\MachineGuid
10:47....	wotsuper1.exe	3544	RegCloseKey	HKLM\Software\Microsoft\Cryptography
10:47....	wotsuper1.exe	3544	RegOpenKey	HKLM\Software\Microsoft\Cryptography\Offload
10:47....	wotsuper1.exe	3544	RegCloseKey	HKLM\SOFTWARE\Microsoft\Cryptography\Defaults\Provider\Microsoft Strong Cryptographic Provider
10:47....	wotsuper1.exe	704	RegOpenKey	HKLM\SYSTEM\CurrentControlSet\Services\crypt32

And also a lot of crypto related registries.

122	67.199962222	192.168.245.130	192.168.245.133	DNS	87 Standard	query 0xb70f A teredo.ipv6.microsoft.com
213	104.749049448	192.168.245.130	192.168.245.133	DNS	78 Standard	query 0x8b2a A go.microsoft.com
227	106.159057834	192.168.245.130	192.168.245.133	DNS	81 Standard	query 0xf86c A manillamemories.com
241	106.296797497	192.168.245.130	192.168.245.133	DNS	74 Standard	query 0x59f8 A iplogger.org
242	106.296933021	192.168.245.130	192.168.245.133	DNS	71 Standard	query 0x97ce A telete.in
351	106.761723752	192.168.245.130	192.168.245.133	DNS	85 Standard	query 0x7ad2 A ctldl.windowsupdate.com
456	110.289952661	192.168.245.130	192.168.245.133	DNS	74 Standard	query 0x33b8 A api.bing.com
457	110.290145192	192.168.245.130	192.168.245.133	DNS	74 Standard	query 0xbc85 A www.bing.com
458	110.290454762	192.168.245.130	192.168.245.133	DNS	74 Standard	query 0x5327 A www.bing.com
813	134.163447296	192.168.245.130	192.168.245.133	DNS	72 Standard	query 0x8dd4 A ip-api.com
852	135.517147631	192.168.245.130	192.168.245.133	DNS	84 Standard	query 0xb6bb A iecvlist.microsoft.com
1046	146.225501619	192.168.245.130	192.168.245.133	DNS	87 Standard	query 0x3413 A teredo.ipv6.microsoft.com
1153	166.711902434	192.168.245.130	192.168.245.133	DNS	71 Standard	query 0xf443 A telete.in
1176	171.321772464	192.168.245.130	192.168.245.133	DNS	84 Standard	query 0xb65b A ieonline.microsoft.com
1189	171.48599744	192.168.245.130	192.168.245.133	DNS	85 Standard	query 0x6153 A ctldl.windowsupdate.com
1224	171.674982232	192.168.245.130	192.168.245.133	DNS	78 Standard	query 0x9663 A go.microsoft.com

This is a look at the DNS requests made. It made requests to manillamemories.com, iplogger.org and ip-api.com. According to static analysis, it shoud have also made a request to wotsuper.com, but that did not happen in the run.

ip.src==192.168.245.130 and tcp						
No.	Time	Source	Destination	Protocol	Length	Info
235	106.209686157	192.168.245.130	192.168.245.133	TCP	62	52669 → 80 [ACK] Seq=487 Ack=151 Win=65536 Len=0
237	106.209866776	192.168.245.130	192.168.245.133	TCP	62	52669 → 80 [ACK] Seq=487 Ack=409 Win=65280 Len=0
238	106.210900364	192.168.245.130	192.168.245.133	TCP	62	52669 → 80 [FIN, ACK] Seq=487 Ack=409 Win=65280 Len=0
240	106.212357445	192.168.245.130	192.168.245.133	TCP	62	52669 → 80 [ACK] Seq=488 Ack=410 Win=65280 Len=0
245	106.319576521	192.168.245.130	192.168.245.133	TCP	68	52670 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
247	106.319810960	192.168.245.130	192.168.245.133	TCP	62	52670 → 80 [ACK] Seq=1 Ack=1 Win=65536 Len=0
248	106.319871794	192.168.245.130	192.168.245.133	HTTP	414	GET /freeb13.dll HTTP/1.1
250	106.320445388	192.168.245.130	192.168.245.133	TCP	68	52671 → 443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
252	106.320612673	192.168.245.130	192.168.245.133	TCP	62	52671 → 443 [ACK] Seq=1 Ack=1 Win=65536 Len=0
253	106.321247233	192.168.245.130	192.168.245.133	TLSv1	177	Client Hello
256	106.325979294	192.168.245.130	192.168.245.133	TCP	62	52671 → 443 [FIN, ACK] Seq=122 Ack=8 Win=65536 Len=0
257	106.326110921	192.168.245.130	192.168.245.133	TCP	68	52672 → 443 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
259	106.326301999	192.168.245.130	192.168.245.133	TCP	62	52672 → 443 [ACK] Seq=1 Ack=1 Win=65536 Len=0
260	106.327996215	192.168.245.130	192.168.245.133	SSLv3	114	Client Hello
263	106.333130671	192.168.245.130	192.168.245.133	TCP	62	52671 → 443 [ACK] Seq=123 Ack=9 Win=65536 Len=0
266	106.342386623	192.168.245.130	192.168.245.133	TCP	62	52670 → 80 [ACK] Seq=359 Ack=151 Win=65536 Len=0
268	106.342549424	192.168.245.130	192.168.245.133	TCP	62	52670 → 80 [ACK] Seq=359 Ack=409 Win=65280 Len=0
269	106.343459715	192.168.245.130	192.168.245.133	TCP	62	52672 → 443 [FIN, ACK] Seq=58 Ack=8 Win=65536 Len=0
271	106.345069974	192.168.245.130	192.168.245.133	TCP	62	52670 → 80 [ACK] Seq=359 Ack=410 Win=65280 Len=0
273	106.346315079	192.168.245.130	192.168.245.133	TCP	62	52672 → 443 [ACK] Seq=60 Ack=9 Win=65536 Len=0
274	106.354822858	192.168.245.130	192.168.245.133	TCP	62	52670 → 80 [FIN, ACK] Seq=359 Ack=410 Win=65280 Len=0
276	106.368083407	192.168.245.130	192.168.245.133	TCP	68	52673 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
278	106.369164895	192.168.245.130	192.168.245.133	TCP	62	52673 → 80 [ACK] Seq=1 Ack=1 Win=65536 Len=0
279	106.381154472	192.168.245.130	192.168.245.133	HTTP	414	GET /mozglue.dll HTTP/1.1
282	106.401107148	192.168.245.130	192.168.245.133	TCP	62	52673 → 80 [ACK] Seq=359 Ack=151 Win=65536 Len=0
284	106.401319816	192.168.245.130	192.168.245.133	TCP	62	52673 → 80 [ACK] Seq=359 Ack=409 Win=65280 Len=0
286	106.403929629	192.168.245.130	192.168.245.133	TCP	62	52673 → 80 [ACK] Seq=359 Ack=410 Win=65280 Len=0
288	106.412517158	192.168.245.130	192.168.245.133	TCP	62	52673 → 80 [FIN, ACK] Seq=359 Ack=410 Win=65280 Len=0
289	106.428493038	192.168.245.130	192.168.245.133	TCP	68	52674 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
291	106.428667556	192.168.245.130	192.168.245.133	TCP	62	52674 → 80 [ACK] Seq=1 Ack=1 Win=65536 Len=0
292	106.428771218	192.168.245.130	192.168.245.133	HTTP	415	GET /msvcpl48.dll HTTP/1.1
295	106.446911648	192.168.245.130	192.168.245.133	TCP	62	52674 → 80 [ACK] Seq=360 Ack=151 Win=65536 Len=0
297	106.447026994	192.168.245.130	192.168.245.133	TCP	62	52674 → 80 [ACK] Seq=360 Ack=409 Win=65280 Len=0
299	106.449481796	192.168.245.130	192.168.245.133	TCP	62	52674 → 80 [ACK] Seq=360 Ack=410 Win=65280 Len=0
300	106.484238475	192.168.245.130	192.168.245.133	TCP	62	52674 → 80 [FIN, ACK] Seq=360 Ack=410 Win=65280 Len=0
302	106.485769255	192.168.245.130	192.168.245.133	TCP	68	52675 → 80 [SYN] Seq=0 Win=8192 Len=0 MSS=1460 WS=256 SACK_PERM=1
1030	141.6842299072	192.168.245.133	192.168.245.130	TCP	56	443 → 52734 [FIN, ACK] Seq=8 Ack=123 Win=64128 Len=0
1032	141.686861486	192.168.245.133	192.168.245.130	TCP	56	443 → 52735 [FIN, ACK] Seq=8 Ack=60 Win=64256 Len=0
1049	146.682059165	192.168.245.130	192.168.245.130	TCP	68	443 → 52736 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=
1052	146.682722589	192.168.245.133	192.168.245.130	TCP	56	443 → 52736 [ACK] Seq=1 Ack=122 Win=64128 Len=0
1053	146.686575592	192.168.245.133	192.168.245.130	TLSv1	63	Alert (Level: Fatal, Description: Protocol Version)
1056	146.687170498	192.168.245.133	192.168.245.130	TCP	68	443 → 52737 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=
1059	146.687685493	192.168.245.133	192.168.245.130	TCP	56	443 → 52737 [ACK] Seq=1 Ack=59 Win=64256 Len=0
1060	146.691000889	192.168.245.133	192.168.245.130	SSLv3	63	Alert (Level: Fatal, Description: Handshake Failure)
1062	146.691903982	192.168.245.133	192.168.245.130	TCP	56	443 → 52736 [FIN, ACK] Seq=8 Ack=123 Win=64128 Len=0
1064	146.695734583	192.168.245.133	192.168.245.130	TCP	56	443 → 52737 [FIN, ACK] Seq=8 Ack=60 Win=64256 Len=0
1070	151.689563226	192.168.245.133	192.168.245.130	TCP	68	443 → 52738 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=
1073	151.690140549	192.168.245.133	192.168.245.130	TCP	56	443 → 52738 [ACK] Seq=1 Ack=122 Win=64128 Len=0
1074	151.693877955	192.168.245.133	192.168.245.130	TLSv1	63	Alert (Level: Fatal, Description: Protocol Version)
1077	151.694391869	192.168.245.133	192.168.245.130	TCP	68	443 → 52739 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=
1080	151.694876577	192.168.245.133	192.168.245.130	TCP	56	443 → 52739 [ACK] Seq=1 Ack=59 Win=64256 Len=0
1081	151.697957934	192.168.245.133	192.168.245.130	SSLv3	63	Alert (Level: Fatal, Description: Handshake Failure)
1083	151.698832804	192.168.245.133	192.168.245.130	TCP	56	443 → 52738 [FIN, ACK] Seq=8 Ack=123 Win=64128 Len=0
1085	151.703069767	192.168.245.133	192.168.245.130	TCP	56	443 → 52739 [FIN, ACK] Seq=8 Ack=60 Win=64256 Len=0
1100	156.697207009	192.168.245.133	192.168.245.130	TCP	68	443 → 52740 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=
1103	156.697840877	192.168.245.133	192.168.245.130	TCP	56	443 → 52740 [ACK] Seq=1 Ack=122 Win=64128 Len=0
1104	156.701998401	192.168.245.133	192.168.245.130	TLSv1	63	Alert (Level: Fatal, Description: Protocol Version)
1107	156.702603516	192.168.245.133	192.168.245.130	TCP	68	443 → 52741 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=

It can also be then seen constantly making multiple TCP and HTTP requests, suggesting communication with its C2 server. These coincide with the existence of informations.txt and passwords.txt, suggesting that its sharing these files with its C2 server. After pressing OK multiple times in the error thrown by wotsuper1 as mentioned above, it deletes all the dll files in ProgramData, and passwords.txt and information.txt are nowhere to be found.

Therefore:

- Querying multiple system registries and system files
- Querying multiple cryptographic registries
- Use and deletion of DLLs like nss3
- Existence and disappearance of information.txt and passwords.txt
- Continuous TCP and HTTP communication with a potential C2 server.

Are strong indicators that is an **Information and Password stealer**.

OFFLINE EXECUTION:

When the malware is run without inetsim, there is no error thrown. the 'gibberish' folder does not delete its contents. Upon unzipping the zipped folder in it, information.txt and passwords.txt is revealed. This is probably because it never reaches the delete file commands when it is not connected to the internet. This made analysis easier.

files	3/28/2025 8:51 PM	File folder	
_63c09915-9fbb-48a0-b874-f23fc25b54d5...	3/28/2025 8:51 PM	Compressed (zipp...	2 KB
information	3/28/2025 8:51 PM	Text Document	3 KB
passwords	3/28/2025 8:51 PM	Text Document	0 KB

Following are the contents of information.txt. It contains all the information related to the system.

```
Version: 28.3
Date: Fri Mar 28 20:51:48 2025
MachineID: 63c09915-9fbb-48a0-b874-f23fc25b54d5
GUID: {e29ac6c0-7037-11de-816d-806e6f6e6963}
HWID: 63c09915-9fbb-48a0-b874-816d-806e6f6e6963

Path: C:\Program Files\wotsuper\wotsuper\wotsuper1.exe
Work Dir: C:\ProgramData\87NELS8D6OK6UCKSABE3CWV3D

windows: windows 7 Professional [x86]
Computer Name: 2022MALWARE7-2
User Name: malware
Display Resolution: 1920x980
Display Language: en-US
Keyboard Languages: English (United States)
Local Time: 28/3/2025 20:51:48
TimeZone: UTC-5

[Hardware]
Processor: AMD EPYC 7402 24-Core Processor
CPU Count: 1
RAM: 2047 MB
VideoCard: VMware SVGA 3D

[Network]
IP:
Country: ()
City: ()
ZIP:
Coordinates: ,
ISP: ()

[Processes]
----- system [4]
----- smss.exe [248]
- csrss.exe [336]
- csrss.exe [388]
- wininit.exe [396]
- winlogon.exe [428]
- services.exe [492]
- lsass.exe [500]
- lsm.exe [508]
- svchost.exe [616]
- svchost.exe [692]
- svchost.exe [776]
- svchost.exe [816]
- svchost.exe [844]
```

Passwords.txt is empty, possibly because the browser does not have any saved passwords.

Wotsuper.exe persists in both offline and online environments, but wotsuper1 stops after pressing 'OK' multiple times in the error dialog box, which is only seen in an online environment (persists in an offline environment).

Indicators of Compromise:

The biggest indicator is the presence of wotsuper folder in Program Files, and the 4 files within it, namely wotsuper.exe, wotsuper1.exe, uninstall.exe, and uninstall.ini.

In both offline and online executions, the 'gibberish' folder remains in ProgramData, though, in online execution, it deletes its contents within.

Requests to the URLs manillamemories.com, iplogger.org and ip-api.com also indicate the presence and execution of this malware.

Yara:

```
remnux@remnux:~/Desktop$ cat detect_wotsuper.yara
rule detect_wotsuper
{
    meta:
        description = "Detects malware sample based on bobsoft and wotsuper"
        author = "Vandan Agrawal"
        date = "2025-03-28"

    strings:
        $wotsuper = "wotsuper" nocase
        $iplogger = "iplogger" nocase

    condition:
        $wotsuper and $iplogger
}
remnux@remnux:~/Desktop$ yara -r detect_wotsuper.yara ./PMA\ Labs/.
remnux@remnux:~/Desktop$ yara detect_wotsuper.yara sample2.exe
detect_wotsuper sample2.exe
remnux@remnux:~/Desktop$
```